Lab01-Objects-Classes

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VNU-HCM, HCMUS, FIT, SE

CS202 - Programming Systems

25A01 - 02

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A: YY: 01

H: YY: 05

Design class diagrams using Mermaid format and implement the required functionality in C++ for the following assignments. For your final submission, please ensure you include the C++ source code, the raw Mermaid syntax, and the rendered image files for each class diagram.

Assignment 1

Class: Fraction

Attributes:

1. Numerator
2. Denominator

Methods:

1. Input
2. Output
3. Add 2 fractions
4. Substract 2 fractions

5. Multiply 2 fractions
6. Divide 2 fractions. throw; if divided by zero
7. Reduce: 2/4 ⇒ 1/2
8. Compare
9. IsPositive
10. IsNegative
11. IsZero

Students must create a `main()` function with a menu to select the demonstration method.

Assignment 2

Class Point.

Attributes:

1. x
2. y

Methods:

1. Input
2. Output
3. Distance from point A to point B
4. Distance to Ox
5. Distance to Oy

Class: Triangle

Attributes:

1. Point A
2. Point B
3. Point C

Methods:

1. Input

2. Output
3. IsValidTriangle
4. Type of a triangle.
5. Perimeter
6. Area
7. Center G

Students must create a `main()` function with a menu to select the demonstration method.

Assignment 3

Write a presentation (Markdown format) with at least 20 slides to explain some OOP topics on Python:

- How to define classes
- How to define attributes
- How to define methods.
- How to call methods.
- OOP scope (private, public).

Assignment 4

Write a presentation (Markdown format) with at least 20 slides to explain some OOP topics on Java:

- How to define classes
- How to define attributes
- How to define methods.
- How to call methods.
- OOP scope (private, public).

Assignment 5

Given the provided C/C++ source code written in a procedural programming style, apply your prompt engineering skills to convert it into an Object-Oriented Programming (OOP) paradigm. Ensure that the converted source code compiles and executes perfectly. Additionally, list all the classes, attributes, and methods present in your new source code. Finally, write a Markdown report documenting your step-by-step process for completing this assignment, including the specific prompts you utilized.